

Лабораторная работа № 14

Настройка файловых служб Samba

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Цель:

Получить навыки настройки группового доступа к общим ресурсам по SMB.



Установка Samba

```
root@server:~  
Complete!  
root@server:~# groupadd -g 1010 sambagroup  
root@server:~# usermod -aG sambagroup ndbaranov  
root@server:~# mkdir -p /srv/sambashare  
root@server:~# nano /etc/samba/smb.conf  
root@server:~# testparm  
Load smb config files from /etc/samba/smb.conf  
Loaded services file OK.  
Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)  
  
Server role: ROLE_STANDALONE  
  
Press enter to see a dump of your service definitions  
  
# Global parameters  
[global]  
    printcap name = cups  
    security = USER  
    workgroup = NDBARANOV-NET  
    idmap config * : backend = tdb  
    cups options = raw  
    include = /etc/samba/usershares.conf  
  
[homes]  
    browseable = No  
    comment = Home Directories  
    inherit acls = Yes  
    read only = No  
    valid users = %S %D%%S  
  
[printers]  
    browseable = No  
    comment = All Printers  
    create mask = 0600  
    path = /var/tmp  
    printable = Yes  
  
[print$]  
    comment = Printer Drivers  
    create mask = 0664  
    directory mask = 0775  
    force group = printadmin  
    path = /var/lib/samba/drivers  
    write list = printadmin root  
  
[sambashare]  
    comment = My Samba Share  
    path = /srv/sambashare  
    write list = @sambagroup  
root@server:~#
```

Смысл результата

На server установлены samba, samba-client и cifs-utils, создана группа sambagroup. Службы SMB подготовлены к запуску.

Учётная запись Samba

```
Выбрать root@server:~
root@server:~# systemctl start smb
systemctl enable smb
systemctl status smb --no-pager
Created symlink '/etc/systemd/system/multi-user.target.wants/smb.service' → '/usr/lib/systemd/system/smb.service'.
● smb.service - Samba SMB Daemon
   Loaded: loaded (/usr/lib/systemd/system/smb.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-09-06 21:52:46 UTC; 707ms ago
 Invocation: f7e91bf9608248bcaeacace29ba95429
    Docs: man:smbd(8)
          man:samba(7)
          man:smb.conf(5)
 Main PID: 13219 (smbd)
   Status: "smbd: ready to serve connections..."
    Tasks: 3 (limit: 10232)
  Memory: 13.4M (peak: 13.7M)
     CPU: 163ms
    CGroup: /system.slice/smb.service
            └─13219 /usr/sbin/smbd --foreground --no-process-group
              └─13222 /usr/sbin/smbd --foreground --no-process-group
                └─13223 /usr/sbin/smbd --foreground --no-process-group

Sep 06 21:52:45 server systemd[1]: Starting smb.service - Samba SMB Daemon...
Sep 06 21:52:46 server systemd[1]: Started smb.service - Samba SMB Daemon.
root@server:~#
```

Смысл результата

Пользователь добавлен в базу smbpasswd и включён для сетевой аутентификации. Пароль Samba хранится отдельно от системного пароля.

Ресурсы sambashare

```
ndbaranov@server/srv/sambashare
root@server:~# smbclient -L //server.ndbaranov.net
Password for [NDBARANOV-NET\root]:
Anonymous login successful

      Sharename      Type            Comment
      -----
      print$         Disk            Printer Drivers
      sambashare      Disk            My Samba Share
      IPC$            IPC             IPC Service (Samba 4.23.5)
SMB1 disabled -- no workgroup available
root@server:~# less /usr/lib/iptables/services/samba.xml
root@server:~# firewall-cmd --add-service=samba
firewall-cmd --add-service=samba --permanent
firewall-cmd --reload
success
success
success
root@server:~# chgrp sambagroup /srv/sambashare
chmod g=rwx /srv/sambashare
root@server:~# cd /srv
ls -Z
unconfined_u:object_r:nfs_t:s0 nfs unconfined_u:object_r:var_t:s0 sambashare
root@server:/srv# semanage fcontext -a -t samba_share_t "/srv/sambashare(/.*)?"
restorecon -vR /srv/sambashare
Relabeled /srv/sambashare from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:samba_share_t:s0
root@server:/srv# cd /srv
ls -Z
unconfined_u:object_r:nfs_t:s0 nfs unconfined_u:object_r:samba_share_t:s0 sambashare
root@server:/srv# setsebool samba_export_all_rw 1
setsebool samba_export_all_rw 1 -P
root@server:/srv# id ndbaranov
uid=1001(ndbaranov) gid=1001(ndbaranov) groups=1001(ndbaranov),10(wheel),1010(sambagroup)
root@server:/srv# su - ndbaranov
Last login: Sun Sep  6 21:39:10 UTC 2026 on pts/1
[ndbaranov@server ~]$ su - ndbaranov
cd /srv/sambashare
touch ndbaranov@server.txt
ls -la
Password:
Last login: Sun Sep  6 21:56:55 UTC 2026 on pts/4
[ndbaranov@server ~]$ cd /srv/sambashare
[ndbaranov@server sambashare]$ touch ndbaranov@server.txt
[ndbaranov@server sambashare]$ ls -la
total 0
drwxrwxr-x. 2 root sambagroup 34 Sep  6 21:57 .
drwxr-xr-x. 4 root root       35 Sep  6 21:50 ..
-rw-r--r--. 1 ndbaranov ndbaranov  0 Sep  6 21:57 ndbaranov@server.txt
[ndbaranov@server sambashare]$
```

Смысл результата

В smb.conf описан общий каталог sambashare с разрешением записи для нужной группы. testparm проверяет синтаксис итоговой конфигурации.

Запуск службы SMB

```
root@client:~  
Install 2 Packages  
Total download size: 850 k  
Installed size: 2.9 M  
Downloading Packages:  
(1/2): cifs-utils-7.6-2.el10_2.x86_64.rpm 828 kB/s | 114 kB 00:00  
(2/2): samba-client-4.23.5-109.el10_2.x86_64.rpm 3.0 MB/s | 736 kB 00:00  
-----  
Total 766 kB/s | 850 kB 00:01  
Running transaction check  
Transaction check succeeded.  
Running transaction test  
Transaction test succeeded.  
Running transaction  
  Preparing      : 1/1  
  Installing     : samba-client-4.23.5-109.el10_2.x86_64 1/2  
  Running scriptlet: samba-client-4.23.5-109.el10_2.x86_64 1/2  
  Installing     : cifs-utils-7.6-2.el10_2.x86_64 2/2  
  Running scriptlet: cifs-utils-7.6-2.el10_2.x86_64 2/2  
  
Installed:  
  cifs-utils-7.6-2.el10_2.x86_64      samba-client-4.23.5-109.el10_2.x86_64  
  
Complete!  
root@client:~# less /usr/lib/firewalld/services/samba-client.xml  
root@client:~# firewall-cmd --add-service=samba-client --permanent  
firewall-cmd --add-service=samba-client --permanent  
firewall-cmd --reload  
success  
success  
success  
root@client:~# groupadd -g 1010 sambagroup  
root@client:~# usermod -aG sambagroup ndbaranov  
root@client:~# nano /etc/samba/smb.conf  
root@client:~# smbclient -L //server.ndbaranov.net  
Password for [NDBARANOV-NET\root]:  
Anonymous login successful  
  
  Sharename      Type      Comment  
  -----      -  
  print$         Disk      Printer Drivers  
  sambashare     Disk      My Samba Share  
  IPC$           IPC       IPC Service (Samba 4.23.5)  
  
SMB1 disabled -- no workgroup available  
root@client:~# smbclient -L //server.ndbaranov.net -U ndbaranov  
Password for [NDBARANOV-NET\ndbaranov]:  
  
  Sharename      Type      Comment  
  -----      -  
  print$         Disk      Printer Drivers  
  sambashare     Disk      My Samba Share  
  IPC$           IPC       IPC Service (Samba 4.23.5)  
  ndbaranov      Disk      Home Directories  
  
SMB1 disabled -- no workgroup available  
root@client:~#
```

Смысл результата

Служба smb включена и запущена. В статусе systemd видно, что конфигурация принята и сервер готов обрабатывать подключения.

Обнаружение общих ресурсов

```
root@client:~  
root@client:~# mkdir -p /mnt/samba  
root@client:~# mount -o username=ndbaranov,user,rw,uid=ndbaranov,gid=sambagroup //server.ndbaranov.net/s  
ambashare /mnt/samba  
Password for ndbaranov@//server.ndbaranov.net/sambashare:  
root@client:~# mount | grep '/mnt/samba'  
//server.ndbaranov.net/sambashare on /mnt/samba type cifs (rw,nosuid,nodev,noexec,relatime,vers=3.1.1,ca  
che=strict,upcall_target=app,username=ndbaranov,uid=1001,forceuid,gid=1010,forcegid,addr=192.168.1.1,fil  
e_mode=0755,dir_mode=0755,soft,nounix,serverino,mapposix,repase=nfs,nativesocket,symlink=native,rsiz=4  
194304,wsiz=4194304,bsiz=1048576,echo_interval=60,actimeo=1,closetimeo=1,user)  
root@client:~#
```

Смысл результата

smbclient -L выводит sambashare в списке доступных ресурсов. Проверка выполнена с учётными данными созданного пользователя.

Выбрать ndbaranov@client:/mnt/samba

```
root@client:~# su - ndbaranov
Last login: Sun Sep  6 21:42:21 UTC 2026 on pts/1
ndbaranov@client:~$ cd /mnt/samba
ndbaranov@client:/mnt/samba$ ls -la
total 0
drwxr-xr-x. 2 ndbaranov sambagroup 0 Sep  6 21:57 .
drwxr-xr-x. 4 root          root      30 Sep  6 22:01 ..
-rwxr-xr-x. 1 ndbaranov sambagroup 0 Sep  6 21:57 ndbaranov@server.txt
ndbaranov@client:/mnt/samba$ touch ndbaranov@client.txt
ndbaranov@client:/mnt/samba$ ls -la
total 0
drwxr-xr-x. 2 ndbaranov sambagroup 0 Sep  6 22:01 .
drwxr-xr-x. 4 root          root      30 Sep  6 22:01 ..
-rwxr-xr-x. 1 ndbaranov sambagroup 0 Sep  6 22:01 ndbaranov@client.txt
-rwxr-xr-x. 1 ndbaranov sambagroup 0 Sep  6 21:57 ndbaranov@server.txt
ndbaranov@client:/mnt/samba$
```

Смысл результата

Каталог получил контекст samba_share_t, а в firewalld разрешён сервис samba. Эти настройки обеспечивают доступ на уровне политики и сети.

Подключение CIFS

```
root@client:~  
root@client:~# umount /mnt/samba  
root@client:~# touch /etc/samba/smbusers  
root@client:~# chmod 600 /etc/samba/smbusers  
root@client:~# nano /etc/samba/smbusers  
root@client:~# ls -l /etc/samba/smbusers/samba/smbusers  
-rw----- 1 root root 35 Sep  6 22:02 /etc/samba/smbusers  
root@client:~# nano /etc/fstab  
root@client:~# mount -a  
mount: (hint) your fstab has been modified, but systemd still uses  
the old version; use 'systemctl daemon-reload' to reload.  
root@client:~# mount | grep '/mnt/samba'  
//server.ndbaranov.net/smbashare on /mnt/samba type cifs (rw,nosuid,nodev,noexec,relatime,vers=3.1.1,cache=strict,upcall_target=app,username=ndbaranov,uid=1001,forceuid,gid=1010,forcegid,addr=192.168.1.1,file_mode=0755,dir_mode=0755,soft,nounix,serverino,mapposix,repase=nfs,nativesocket,symlink=native,rsize=4194304,wsz=4194304,bsize=1048576,echo_interval=60,actimeo=1,closetimeo=1,user,_netdev)  
root@client:~# ls -la /mnt/samba  
total 0  
drwxr-xr-x. 2 ndbaranov sambagroup  0 Sep  6 22:01 .  
drwxr-xr-x. 4 root      root      30 Sep  6 22:01 ..  
-rwxr-xr-x. 1 ndbaranov sambagroup  0 Sep  6 22:01 ndbaranov@client.txt  
-rwxr-xr-x. 1 ndbaranov sambagroup  0 Sep  6 21:57 ndbaranov@server.txt  
root@client:~# systemctl daemon-reload  
root@client:~# mount -a  
root@client:~# mount | grep '/mnt/samba'  
//server.ndbaranov.net/smbashare on /mnt/samba type cifs (rw,nosuid,nodev,noexec,relatime,vers=3.1.1,cache=strict,upcall_target=app,username=ndbaranov,uid=1001,forceuid,gid=1010,forcegid,addr=192.168.1.1,file_mode=0755,dir_mode=0755,soft,nounix,serverino,mapposix,repase=nfs,nativesocket,symlink=native,rsize=4194304,wsz=4194304,bsize=1048576,echo_interval=60,actimeo=1,closetimeo=1,user,_netdev)  
root@client:~# ls -la /mnt/samba  
total 0  
drwxr-xr-x. 2 ndbaranov sambagroup  0 Sep  6 22:01 .  
drwxr-xr-x. 4 root      root      30 Sep  6 22:01 ..  
-rwxr-xr-x. 1 ndbaranov sambagroup  0 Sep  6 22:01 ndbaranov@client.txt  
-rwxr-xr-x. 1 ndbaranov sambagroup  0 Sep  6 21:57 ndbaranov@server.txt  
root@client:~#
```

Смысл результата

Клиент смонтировал
//server.ndbaranov.net/smbashare в
/mnt/samba. Список монтирований
показывает файловую систему типа cifs.

Проверка записи

```
PowerShell 7 (x64)
server: Loaded services file OK.
server: Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)
server:
server: Server role: ROLE_STANDALONE
server:
server: # Global parameters
server: [global]
server:   printcap name = cups
server:   security = USER
server:   workgroup = NDBARANOV-MET
server:   idmap config * : backend = tdb
server:   cups options = raw
server:   include = /etc/samba/usershares.conf
server:
server: [homes]
server:   browseable = No
server:   comment = Home Directories
server:   inherit acls = Yes
server:   read only = No
server:   valid users = %S %D%%S
server:
server: [printers]
server:   browseable = No
server:   comment = All Printers
server:   create mask = 0600
server:   path = /var/tmp
server:   printable = Yes
server:
server: [print$]
server:   comment = Printer Drivers
server:   create mask = 0664
server:   directory mask = 0775
server:   force group = printadmin
server:   path = /var/lib/samba/drivers
server:   write list = printadmin root
server:
server: [smbashare]
server:   comment = My Samba Share
server:   path = /srv/smbashare
server:   write list = @smbagroup
server: Configure Samba user
server: Make share dir
server: Configure SELinux
server: File context for /srv/smbashare(/.*)? already defined, modifying instead
server: Configure firewall
server: Warning: ALREADY_ENABLED: samba
server: success
server: success
server: Start Samba
server: SMB server provisioning completed
PS D:\work\ndbaranov\vagrant>
```

Смысл результата

Через подключённый ресурс создан тестовый файл; он отображается и на сервере. Это подтверждает права группы на запись.

Credentials, fstab и provision

```
PS D:\work\ndbaranov\vagrant> vagrant provision client --provision-with "SMB client"
==> client: Running provisioner: SMB client (shell)...
client: Running: C:/Users/barash/AppData/Local/Temp/vagrant-shell120260907-23336-vydagb.sh
client: Provisioning SMB client
client: Install needed packages
client: Last metadata expiration check: 0:47:21 ago on Sun 06 Sep 2026 09:20:22 PM UTC.
client: Package samba-client-4.23.5-109.el10_2.x86_64 is already installed.
client: Package cifs-utils-7.6-2.el10_2.x86_64 is already installed.
client: Dependencies resolved.
client: Nothing to do.
client: Complete!
client: Users and groups
client: Configure Samba
client: Configure credentials
client: Configure firewall
client: Warning: ALREADY_ENABLED: samba-client
client: success
client: success
client: Configure mount
client: SMB client provisioning completed
```

Смысл результата

Учётные данные вынесены в защищённый файл, монтирование добавлено в fstab, затем выполнены provision-сценарии Samba. Настройка готова к автоматическому восстановлению.

Результат

Samba предоставляет общий ресурс по SMB с групповой аутентификацией. CIFS-монтирование и создание файла подтверждают сетевую доступность и права на запись.